Nortek Vector PUV Pipeline

User Guide and Reference

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Abstract

This guide is for anyone running the Nortek Vector PUV pipeline on their own data or using the San Diego catalog it produced. The pipeline takes raw pressure–velocity (PUV) instrument files through four built levels into quality-controlled time series, hourly wave spectra, wave-forcing metrics and infragravity diagnostics, driven by one config file per deployment. Sections 2–5 cover installing, configuring and running a deployment. Section 6 covers the per-channel quality control and the flags every segment carries, and Section 7 the provenance stamps and the published catalog. Sections 8–10 are the methods and output reference. To run the code, read Sections 2–7. The repository `README.md` is kept current with every code change and is the place to check if this guide and the code ever disagree; `CHANGELOG.md` lists the changes.

Contents

1 What the pipeline is **2**

2 Getting started **2**
2.1 Requirements . . . 2
2.2 First run . . . 3

3 Configuring a deployment **4**
3.1 Adding a new deployment . . . 4
3.2 Quality-control options: `cfg.qcOpts` . . . 5

4 Raw-binary ingestion and the pre-2019 archive **5**
4.1 Choosing the ingest path . . . 6
4.2 Two things the firmware-1.21 recorders get wrong . . . 6
4.3 Robustness the archive forced . . . 6
4.4 Split recorder files . . . 7

5 Running the pipeline, level by level **7**

6 Quality control and segment flags **8**
6.1 Why channel decoupling . . . 8
6.2 Per-channel validity . . . 8
6.3 The temperature test . . . 8
6.4 Sound-speed correction . . . 8
6.5 Segment flags . . . 9

6.6 Failure signatures 9

7 Provenance and the published catalog **10**

8 Methods **10**
8.1 L1: quality control and coordinate rotation 10
8.2 L2: spectral analysis 11
8.3 L3: wave forcing characterization 12
8.4 L4: nonlinear and infragravity diagnostics 12

9 Validation **13**

10 Output reference **13**

11 Known limitations **14**

12 Troubleshooting and where to look next **14**

1 What the pipeline is

Bottom-mounted Nortek Vector acoustic Doppler velocimeters (ADVs) record pressure and three-component velocity continuously at 2 Hz. The pipeline reads either form of Nortek raw data, the ExploreV ASCII export (`.dat`/`.sen`/`.hdr`) or the recorder binary (`.VEC`/`.vec`/`.049`), which it decodes itself (§4), so neither ExploreV nor Windows is needed. Every deployment runs through the same drivers and produces the same output structures. The San Diego catalog built with it holds 65 instrument-records from 46 deployments, 2014–2026, at Torrey Pines, Los Peñasquitos, Solana Beach, Cardiff, SIO Pier, Imperial Beach, Coronado and Catalina Island, at nominal depths of 5 to 30 m. It is published on the SIO group share as catalog version 2026-10-14.1.

Level	Output	What it contains
L1	QC'd 2 Hz time series	Per-file bursts, clock drift and offset, tilt correction, per-channel quality control, sound-speed correction, rotation to a geographic frame
L2	Hourly segments	Multitaper spectra, pressure correction to surface elevation, bulk wave parameters (H_s, T_p, direction), near-bed velocity, bed and Reynolds stress, velocity moments, pressure–velocity consistency diagnostics
L3	Forcing metrics	Band energy (infragravity, swell, sea), storm events, transport proxies (Shields, settling velocity, Rouse, mobilization), tidal and subtidal currents
L4	Nonlinear / IG diagnostics	Surface elevation in three bands, incident/reflected infragravity split, bound/free infragravity separation, bispectra, frequency-resolved moments, velocity PDFs; `L4_xspec` holds pairwise infragravity cross-spectra for multi-instrument deployments
L5	*(planned)*	PUV–altimeter merge: bed change against forcing

L1–L4 need no altimeter data. Each level reads the previous level's `.mat` files, so the levels run in order L1→L2→L3→L4.

2 Getting started

2.1 Requirements

- **MATLAB R2025a**, the release used for development and testing; earlier releases are untested. Required toolboxes: **Signal Processing** (DPSS multitaper spectra, `cpsd`) and **Aerospace** (`igrfmagm`, the IGRF magnetic declination applied at L1). The **Parallel Computing Toolbox** is optional; it speeds up `PUV_L4_bispectra`, and the serial path gives bit-identical results. The Mapping and Statistics toolboxes are not needed by L1–L4.

- **Files outside the repository, for L3 only.** L1, L2 and L4 use only files in the repository. L3 needs 11 outside files, listed with their SHA-256 in `shared/external_deps.m`: `t_tide` version 1.3 (Pawlowicz, Beardsley and Lentz, 2002; eight files, not redistributed because its license is informal) for the tidal fit, and the SIO Coastal Processes lab toolbox function `read_MOPline2.m` with its helpers `get_k.m` and `xyRotate.m` for the CDIP MOP storm reference. Without `t_tide`, L3 stops with an error unless `opts.allowNoTides = true` is passed;

without the lab toolbox, L3 runs without the MOP storm reference and records the fallback. `README.md`, "External dependencies", gives the fallback folders the code searches.

- **Network access** at runtime for: the CDIP shore-normal angle when a config uses `instr.mopStation`; the CDIP offshore reference (`instr.refStation`); and NOAA tide predictions at L3 (§8.3). A failed NOAA download is an error unless `opts.allowTtideFallback = true`.
- **Raw data**: point `cfg.rawDataRoot` at any folder. The bundled San Diego configs read `/Volumes/group/PUV_data/Vector/`, with deployment metadata in `/Volumes/group/DeploymentNotes/`.

2.2 First run

```
% 1. From the repo root, put the pipeline on the MATLAB path
%    (startup_puv leaves outputs/, raw_cache/, .git/ and .claude/ off it):
cd /path/to/PUV_Pipeline
startup_puv

% 2. Optional: cache the raw files locally. Network reads are slow, and the
%    L1 driver reads raw_cache/<deployment>/ when it exists.
reg = deployment_registry();
f = reg('TBR23');
cfg = f();
copy_raw_to_local(cfg);

% 3. Run the levels in order. Each driver has deployment_name = '...' at the top.
PUV_L1_driver        % raw -> QC'd time series   (outputs/L1/TBR23/)
PUV_L2_driver        % spectra + bulk params     (outputs/L2/TBR23/)
PUV_L3_driver        % forcing characterization  (outputs/L3/TBR23/)
PUV_L4_driver        % nonlinear / IG dynamics   (outputs/L4/TBR23/)
PUV_L4_xspec_driver  % pairwise IG cross-spectra (two or more instruments)
```

The batch scripts `PUV_L1_run_all` ... `PUV_L4_run_all` and `PUV_L4_xspec_run_all` process every registered deployment and skip records whose output already exists. `PUV_L4_run_all` leaves out `bispectra`, the slow module, by default (`runBispectra = false`) and says so at the end. Output files, each with a `.prov.json` sidecar (§7):

- `outputs/L1/<deployment>/<label>_processed.mat`
- `outputs/L{2,3,4}/<deployment>/<label>_L{2,3,4}.mat`
- `outputs/L4_xspec/<deployment>/xspec.mat`

Outputs are not version-controlled; they are reproducible from raw data plus code.

To check an installation, run the self-contained tests:

```
run_all_tests                          % 23 tests: PASS / FAIL / SKIP and a total
run_all_tests('includeData', true)     % also tests that read outputs/, the server or
    the network
```

From a shell use plain `matlab -batch "startup_puv; run_all_tests"`: on R2025a under macOS, adding `-nodisplay` makes `matlab -batch` exit 0 even when the command errors.

3 Configuring a deployment

`config/deployment_registry.m` maps a short name to a config function, for example `'TBR23'` → `@TBR23_config`; 50 names are registered. Names follow `SITE + year + optional season letter` (`TOR24S` = Torrey spring 2024; `SIO24A`–`C` = SIO Pier 2024 in order). `TBR23` (Torrey Pines Beach Recovery 2023) is the one legacy exception, and `TOR23S` is registered as an alias of it. For the pre-2019 seasons, which ran one instrument at a fixed station and swapped it every 5–6 weeks, the trailing letter is the deployment's ordinal within that winter (`TOR15A` = first Torrey deployment of 2015–16).

3.1 Adding a new deployment

The full walkthrough is `docs/NEW_DEPLOYMENT.md`. In outline:

1. Copy `config/TEMPLATE_config.m` to `config/<NAME>_config.m` and rename the function.
2. Per instrument set `label`, `filePrefix`, `rawSubfolder`, `latlon`, `depth_nominal`, `doffp` (pressure-port height above the bed, m; required), `heading` and a shore-normal source. San Diego values come from `/Volumes/group/DeploymentNotes/*.xls`; `config/DOFFP_LOOKUP_CHECKLIST.md` lists where to look and the traps (rows labeled only "Redeploy", `doffp` quoted to the altimeter face).
3. Put the **source of every `doffp`, `latlon`, `heading` and `shorenormal` on the same line** as a comment: workbook, sheet, row, serial number and the cell's wording. `validation/audit_config_provenance.m` lists values without one. A surveyed GPS fix carries 5–7 decimal places; a coordinate at 3 decimal places (about 111 m) was placed by hand.
4. Set `cfg.qcOpts.Tvalid` to the site's water-temperature range (§3.2) and `cfg.tideStation` to the nearest NOAA CO-OPS harmonic station (§8.3).
5. An instrument that failed in the field stays in the config with `instruments(k).knownFailed = '<cause>; <source>'`, so the drivers skip it (§5).
6. Register the deployment, then run `validation/audit_registry_loads.m`. It is the one check that does not swallow a config that throws an error, and it lists configured instruments with no L1 as known failures or missing.

Heading. Leave `heading = NaN` to use the median of the onboard compass, unless the compass drifts or steps while the tilt stays constant; then use the mount heading from the field log. A record whose logged and compass headings differ by more than 10° and that keeps the compass value goes in `shared/heading_disputed_records.m` (12 records at present).

Shore-normal. `instr.shorenormal = <deg>` (offshore-pointing bearing, any site; takes precedence) or `instr.mopStation = 'D0580'` (fetched from CDIP, California). With neither, velocity stays in buoy coordinates and the L4 incident/reflected split is skipped.

Pre-2019 recorders. For a deployment ingested from raw binary (§4) set `instr.rawFormat = 'VEC'`. For a firmware-1.21 recorder with a wrong clock epoch add `instr.clockSource = 'filename'`, `instr.deployYear`, `instr.clockOffsetHours` (§4) and `cfg.qcOpts.cutoffGapSec = 60`. `config/TorreyOffshore_config.m` is a worked example.

3.2 Quality-control options: `cfg.qcOpts`

L1 quality control reads one option struct. The defaults live in `L1_raw_to_qc/l1_qc_opts.m`, overlaid field by field with `cfg.qcOpts`; the result is stored in each L1 file at `PUV.qc.opts`, and `validation/audit_L1_qcopts.m` compares the stored options with the configs.

Field	Default	Meaning
`corrMin`	70	Minimum beam correlation (%), Nortek's recommendation.
`Tvalid`	`[-2 40]`	**Plausible water-temperature range (°C).** The primary thermistor-failure test. See the boxed note.
`maxJump`	`Inf`	Optional per-sample temperature-step gate; `Inf` = off.
`tiltStdMax`	2	Moving-window standard deviation of pitch or roll above which the tilt counts as unstable (°).
`tiltAbsMax`	30	Absolute tilt = toppled frame; always invalidates velocity (°).
`tiltWindow`	120	Samples in the moving window (60 s at 2 Hz).
`cFactorTol`	0.002	Below this fractional sound-speed difference no rescale is applied.
`cutoffGapSec`	1	Gap between consecutive sensor records that counts as a battery cutoff (s). 60 for the firmware-1.21 recorders, which leave a benign few-second gap in nearly every hourly file. Read by `PUV_raw_process`.
`offBedDbar`	absent (off)	Vertical-transit pressure test (dbar). Off in every config; it stays off until the next deliberate L1 rebuild.

Set `Tvalid` to the site's water-temperature range. The default `[-2 40]` catches gross failures (freezing, firmware garbage) anywhere, but it accepts an implausibly cold reading from a failing thermistor. Every San Diego and Catalina config sets `cfg.qcOpts.Tvalid = [9 26]`; a December reading of 8 °C at Torrey Pines is not real water. L1 uses it for the thermistor test and L3 for the per-segment viscosity. Section 6.3 explains why a range test is safe for good data.

4 Raw-binary ingestion and the pre-2019 archive

Most deployments before 2019 exist only as recorder binary (`.VEC`/`.vec`/`.049`) with no ExploreV export, and where an export exists it can be partial: `TORREY02_1.dat` covers 5.1 days of a 174-day record and ends mid-line. The pipeline therefore decodes the binary with `L1_raw_to_qc/read_VEC.m`. The decoder matches the one ASCII export that overlaps it: over 886,319 velocity and 443,131 system records, every integer-valued column (velocity counts, amplitude, correlation, clock, flags) is identical, and pressure and attitude agree to the ASCII's print precision (`test_read_VEC.m`).

4.1 Choosing the ingest path

Set `instr.rawFormat` to `'VEC'` or `'ASCII'` to force a path. Left unset, L1 prefers the ASCII export when a `.dat` exists, which is wrong when an export was interrupted, so L1 warns whenever both forms are present. Sampling rate and coordinate system come from the binary's User Configuration record, so a deployment with a 0-byte `.hdr` still ingests.

4.2 Two things the firmware-1.21 recorders get wrong

Sample rate. The configuration field implies 8 Hz on this firmware, but the records are 2 Hz. `read_VEC` measures the rate from the ratio of velocity to 1 Hz system records and warns when it disagrees with the configuration. Three checks agree: the decoded record ratio is 2, the system-record count equals the clock span in seconds, and the field checkout sheet states "Sample rate = 2Hz, Samples per block = 7168, Block time = 3600 s". `instr.decimateTo` handles a genuine 8 Hz record.

Clock epoch and time zone. These recorders stamp their first sample with a wrong epoch (2000-01-01 or 2002-01-01). The clock runs correctly from that wrong start, and the recorder names each hourly file `MMDDHHMM` for the hour it was set to. With `instr.clockSource = 'filename'` and `instr.deployYear`, `vec_clock_from_filenames.m` recovers one constant offset, the median over every file, and rejects it (on the 98th percentile) unless it is nearly constant, so a wrong filename convention fails loudly. Against the one instrument whose clock was set correctly, the filename-derived time agrees to 33 s.

The recovered clock is **Pacific local time, not UTC**. Cross-correlating L2 depth with the NOAA tide prediction placed every filename-clocked record at a lag of -7 or -8 h, so `instr.clockOffsetHours` is added at L1: 7 h for the 2014 records, 8 h for the later ones. It is a fixed offset, not a time-zone conversion; records entirely in daylight time still want 8 h. Derive the sign rather than fitting it: if the data are $D(t) = W(t + \delta)$ and the prediction is $P(t) = W(t)$, the best lag is $L = -\delta$, so a lag of -8 h means the labels are slow and 8 h is *added*. Applying -8 doubles the error and still looks like a match in a lag scan. `scripts/audit_clock_lag.m` checks every filename-clocked record at lag 0.

4.3 Robustness the archive forced

- **Benign hourly gaps.** Firmware 1.21 drops a few seconds in nearly every hourly file; `cutoffGapSec = 60` keeps them from being read as battery death.
- **Power-on glitch.** Some recorders write one sample at power-on, then a long gap, then the deployment (TOR16B: one ping, then 2463 s). The leading sample is trimmed rather than ending the record.
- **Battery-death clock drift.** A dying battery pulls the clock off in the last hours; the record is truncated at that jump.
- **Year-crossing deployments.** `MMDDHHMM` names sort January ahead of the previous November, so files are ordered by their clock, not by name.

4.4 Split recorder files

The recorder splits large records at 100 MiB with a `_N` suffix. Each raw file is treated as its own burst and the battery-cutoff test runs within each burst, so a seam between files is never read as a gap. Concatenating the files first would let one lost second at a seam truncate months of good data.

5 Running the pipeline, level by level

Each level has a driver (`PUV_L{n}_driver.m`) with a `deployment_name` variable at the top, and a batch version (`PUV_L{n}_run_all.m`) that loops over the registry.

L1: raw to QC'd time series. Reads the raw files (ASCII or binary), corrects clock drift and offset, applies per-channel QC (§6), corrects tilt and sound speed, and rotates velocity to the buoy frame ($+x$ west, $+y$ north, $+z$ up). Writes `<label>_processed.mat` holding a `PUV` struct and its `PUV.qc` sub-struct.

L2: spectra and wave statistics. Cuts the L1 series into 1-hour segments, estimates multitaper spectra, converts pressure to surface elevation, and computes bulk wave parameters, near-bed velocity, bed and Reynolds stress and velocity moments. Most analyses read this level.

L3: forcing characterization. Band energy, storm events (with a CDIP MOP reference for PUV gaps), transport proxies, and tidal and subtidal currents.

L4: nonlinear and IG dynamics. Band-passed surface elevation, the incident/reflected and bound/free infragravity splits, bispectra, frequency-resolved moments and velocity PDFs. `PUV_L4_xspec_driver` adds pairwise infragravity cross-spectra for deployments with two or more instruments; it is the only product that reads `LATLON`, and it must be rebuilt whenever any member L4 changes.

Failures. Every driver keeps going when one record fails, then ends with an error naming each config that threw and each record it did not write, so `matlab -batch` exits nonzero. Seven of the 72 configured records have no L1 because the instrument failed in the field (SIO24B, SIO24C, SIO25A, SOL23/MOP651_5m, TOR24S/MOP586_15m, TOR24W/MOP586_5m, TOR25S/MOP586_5m). Their configs carry `knownFailed = '<cause>; <source>'`, and the drivers skip them with a `KNOWN FAILED` line, so an error at the end means a new failure.

Rebuild order. L3 and L4 are built against L2 as it stood at build time. After any L1 or L2 rerun, rebuild L3 and L4 (and `L4_xspec`), and never index an L4 per-segment array with an L2 segment index: match on time with `shared/l4_l2_index_map.m`. Equal segment counts do not prove alignment, and MATLAB silently accepts a logical mask shorter than the array it indexes. A script that recomputes part of an L4 must save the complete struct.

Audits. After changing configs or rebuilding, run `validation/audit_registry_loads.m` (every config builds), `validation/audit_L1_qcopts.m` (stored L1 options match the configs), `validation/audit_L4_coverage.m` (every L4 complete, aligned to its L2 and consistent with it), `validation/audit_config_provenance.m` (geometry values without a source) and `scripts/audit_clock_lag.m` (filename-clocked records at lag 0). These read `outputs/`.

6 Quality control and segment flags

The guiding principle is **channel decoupling**: pressure, velocity, temperature and tilt are judged independently, and one dead channel never discards data from a channel that is still good.

6.1 Why channel decoupling

A Nortek Vector has a Doppler velocity sensor and an auxiliary sensor block (pressure, thermistor, compass, tilt), and the two fail independently. The original QC nulled an entire data row whenever any channel failed. At Torrey Pines in December 2023 (TOR23W/MOP586_10m), one frame's pressure sensor and thermistor died during a storm while its Doppler channel stayed healthy (beam correlations 92–97%); the old QC discarded the velocity anyway, and 120 hours of storm-peak velocity were recovered by the per-channel rework of July 2026 (`docs/L1_sensor_block_failure_2026-07-09.md`).

6.2 Per-channel validity

`L1_raw_to_qc/puv_channel_qc.m` computes per-sample masks:

- **Velocity** is valid when the minimum beam correlation is at least `corrMin`, the frame is not toppled, and the tilt is stable (below). A dead pressure sensor or thermistor does not invalidate velocity.
- **Pressure** is valid inside $[p_{\mathrm{ref}}/2,\ 2p_{\mathrm{ref}}]$, where p_{ref} is the median of the in-water samples (above 0.5 dbar) in the first 10% of the record, at least 1000 samples. A reference taken early survives a sensor that saturates later, which a whole-record median does not.
- **Temperature** is valid inside `Tvalid`, optionally with the step gate `maxJump`.
- **Tilt** is judged two ways: a moving standard deviation above `tiltStdMax` is an unstable reading; an absolute tilt beyond `tiltAbsMax` means the frame toppled or moved.

Tilt describes the measurement geometry, so it is not treated like the other auxiliary channels. An absolute tilt beyond 30° always invalidates velocity, whatever the state of any other channel. A variability spike invalidates velocity only when the sensor block is trusted; when the block has failed, its jittery tilt reading is discarded and a static tilt from the healthy part of the record is used for the rotation instead.

6.3 The temperature test

The instrument computes sound speed from its measured temperature and uses it to scale velocity, so a failed thermistor biases velocity. The test for "failed" is physical plausibility (`Tvalid`), not distance from the record median. A distance test would flag a genuine cold-upwelling event and corrupt its velocity through the rescale; a range test does not, because a real seawater temperature is valid whatever the rest of the record did.

6.4 Sound-speed correction

Where the instrument's sound speed differs from the one implied by a valid temperature by more than `cFactorTol`, velocity is rescaled following the Nortek manual (N3015-030, §2.4.9):

$$V_{\mathrm{corrected}} = V_{\mathrm{recorded}} \frac{c_{\mathrm{true}}}{c_{\mathrm{measured}}}.$$

The reference sound speed comes from a companion frame if the config supplies one (`cfg.soundSpeedRef`), otherwise from the instrument's own healthy temperature–sound-speed relation at a reference temperature taken from in-water, plausible samples, never extrapolated. On healthy data the factor is exactly 1.

6.5 Segment flags

Nothing reconstructed or rescaled is presented as a clean measurement. Every L2 segment carries:

Field	Meaning
`segValid`	All channels good (the original meaning).
`segValid_vel`	Velocity products (moments, mean flow, Reynolds stress) usable.
`segValid_p`	Pressure products (H_s, S_η, depth) usable.
`qc_flag`	1 good, 2 not evaluated, 3 suspect (recovered or sound-speed rescaled), 4 fail (physically implausible). Anything reconstructed is 3, never 1.
`Hs_source`	`'measured'`, `'reconstructed'` or `'none'`.
`vel_c_factor`	The sound-speed factor (exactly 1 on healthy data).
`vel_c_corrected`	Whether the sound-speed rescale was applied.
`qc_record`	Record-level Z flag (pressure against velocity-predicted pressure): `ok`, `FLAG` or `insufficient`. It marks a record and drops nothing.

For clean-only forcing, filter on `qc_flag` = 1. To use storm-peak velocity recovered from a sensor-block failure, take `qc_flag` = 3 with `segValid_vel` true and carry the few-percent scale uncertainty of the rescale. A segment whose pressure is present but implausible (H_s/h above 1.5, or depth more than 50% from nominal) is `qc_flag` = 4.

Two lists apply at reporting time and drop nothing from the files: `shared/excluded_records.m` (RUBY22/MOP582_30m, whose pressure sensor was deployed below its rated depth; RUBY22/MOP579_6m) and `shared/heading_disputed_records.m`, whose directional quantities are uncertain by at least 10°.

6.6 Failure signatures

Three failure modes appear in the archive. A scan of the `.sen` files, which carry battery, sound speed, heading and tilt, separates the first two without reading the much larger `.dat`:

Signature	How to recognize it	Velocity recoverable?
Sensor-block failure	Temperature implausible, sound speed steps, battery erratic; tilt stays small	Yes (with the rescale)
Toppled or moved frame	Temperature, sound speed, battery normal; tilt passes 30° or drifts steadily	No
Doppler failure	Beam correlation collapses to about 5–10%, amplitude drops; the auxiliary block looks fine	No

Beam correlation lives in the `.dat`, so the `.sen` scan cannot see a Doppler failure.

7 Provenance and the published catalog

Sidecars. Every output written by the drivers or the catalog rebuild carries a provenance block, embedded as a `prov` field and written beside the file as `<file>.prov.json` (`shared/save_with_provenance.m`, `build_provenance.m`). It records a `build_id`, the git commit and whether the tree was dirty (`'unknown (no git)'` in a checkout without `.git`), where each key function resolved, the parents' `build_id`s, the external files with their SHA-256, the physical choices applied (for L3: D_{50} and its source, settling-velocity constants, viscosity model, `Tvalid`, tide station and tidal source), any fallback taken, and the SHA-256 of the saved bytes. The drivers have stamped since 2026-10-15. In catalog 2026-10-14.1 the 65 L2, 65 L3, 65 L4 and 9 `L4_xspec` sidecars were written at build time and the 65 L1 sidecars were backfilled (their `backfilled` field says so).

Server catalog. `/Volumes/group/PUV_data/Vector/Processed_HLB/`, laid out `<DEPLOYMENT>/Level{1,2,3,4}_QC/<label>_*.mat` plus `L4_xspec`, each file with its sidecar. Beside them: `manifest.csv` (size, `sha256`, `build_id`, `git_hash`, `backfilled`, `catalog_version_first` per file), every previous manifest, a server `CHANGELOG.md` per catalog version, and a generated `README.md`. Catalog versions are `YYYY-MM-DD.N`. To tell which version a copy belongs to, look up its SHA-256 in the manifest.

Publishing. `scripts/copy_to_server.m` checks every file with `shared/check_prov.m` before copying. It refuses outright a file with no provenance, bytes that differ from the sidecar, a config or derived value that no longer matches (including the tide station), or a parent rebuilt since. It refuses, until a change entry in `docs/catalog_changes/` acknowledges it, a file whose level's code changed since the build, an external dependency that changed, a recorded fallback, or uncommitted code. An entry can be acknowledgement-only (`changes: false`) when the code changed but no output value did. Rule and format: `docs/catalog_changes/README.md`.

No writes into `outputs/` from elsewhere. Only this pipeline writes `outputs/`. An edit from another repository, a notebook or by hand makes a file's bytes disagree with its sidecar, and the push refuses it. To change an output, change the code or config here and rebuild.

8 Methods

A longer methods text, with the history of each change, is in `docs/compile_sections.pdf`.

8.1 L1: quality control and coordinate rotation

Each raw file is read as one burst (ASCII or binary, §4). Clock drift measured at recovery is removed linearly over the deployment, and `clockOffsetHours` is added for filename-clocked records. A gap between sensor records longer than `cutoffGapSec` marks battery depletion and everything after the first such gap is discarded; a power-on glitch at the start is trimmed instead. Per-channel QC (§6) and the sound-speed correction follow.

Where the median tilt exceeds 1°, velocity is rotated to the local vertical sample by sample with the measured pitch ϕ and roll θ. The code applies roll first, then pitch:

$$\mathbf{v}_{\text{corrected}} = \mathbf{R}_{\text{pitch}}(\phi)\,\mathbf{R}_{\text{roll}}(\theta)\,\mathbf{v}_{\text{raw}},$$

with $\mathbf{R}_{\text{roll}}$ a rotation about x and $\mathbf{R}_{\text{pitch}}$ about y. Revisions of this guide before October 2026 wrote the product in the other order; the components in `PUV_raw_process.m` are those of the order above. Whether that order matches Nortek's pitch and roll convention has not been checked.

Velocity is then rotated to the buoy frame ($+x$ west, $+y$ north, $+z$ up; left-handed, the CDIP MOP convention). XYZ records use the heading (the compass median, or the logged mount heading where the compass drifts; §3) plus the IGRF magnetic declination at the deployment position and start date (`igrfmagm`, IGRF-14 where available, else IGRF-13 capped at 2025.0). Records stored in Nortek ENU coordinates, which are magnetic, have the same declination added (`enu_to_buoy_frame.m`, since 2026-10-07; about 11.07° at San Diego). `PUV.rotation.coordSystem` and `.declinationApplied` record which path ran.

8.2 L2: spectral analysis

Segmentation. The L1 series is cut into non-overlapping 1-hour segments of 7200 samples starting at the first UTC top of the hour, the cadence of the CDIP MOP hindcast. Random Doppler error in a segment mean falls as $1/\sqrt{N}$, so the 7200-sample mean has a noise floor $\sqrt{7200/2048} \approx 1.9$ times lower than the 2048-sample (17-minute) segments used before May 2026. Velocity and pressure products are kept separately when no more than 10% of that channel is NaN. Remaining gaps are interpolated linearly in the interior and held at the nearest valid value at a segment edge (since 2026-10-09; before, edge gaps were extrapolated, which inflated infragravity and velocity statistics). Velocity is rotated to shore-normal ($+x$ onshore, $+y$ alongshore north) with `instr.shorenormal` or the CDIP angle for `instr.mopStation`. Depth is $H = \overline{P} \times 10^4/(\rho g) + z_s$ with $\rho = 1025$ kg m^{-3}, $g = 9.81$ m s^{-2}, $\overline{P}$ in dbar, and z_s the pressure-port height above the bed (`doffp`).

Spectral estimation. Spectra use the multitaper method of Thomson (1982) on the whole segment with $NW = 4$ (`shared/psd_multitaper.m`). The code keeps the Slepian tapers whose concentration eigenvalue exceeds 0.99, which for $NW = 4$ is $K = 6$ (the seventh has eigenvalue 0.937), so each bin has about 12 degrees of freedom. Bins are spaced $\Delta f = 2.78 \times 10^{-4}$ Hz and the resolution bandwidth is $2NW/T = 2.2 \times 10^{-3}$ Hz. Auto- and cross-spectra use the same tapers, which keeps the directional moments consistent. Earlier revisions of this guide said $K = 7$; the code has kept six since it was written. Welch (`opts.spectralMethod = 'welch'`) remains available; the comparison that chose multitaper is `docs/multitaper_writeup.pdf`.

Pressure correction. $K_p(f) = \cosh(kz_s)/\cosh(kH)$, with $k(f)$ from $\omega^2 = gk\tanh(kH)$ by Newton iteration, and $S_\eta = S_p/K_p^2$. Bins with $K_p < 0.1$ are set to zero; the cutoff f_{cut} is stored.

Bulk parameters. $H_s = 4\sqrt{m_0}$, $m_0 = \int S_\eta\, df$, over the total (0.004–0.25 Hz), sea–swell (0.04–0.25 Hz) and infragravity (0.004–0.04 Hz) bands. T_p is the period of the sea–swell spectral maximum. $T_{m02} = \sqrt{m_0/m_2}$ and the energy flux $F = \rho g \int c_g S_\eta\, df$ use the total band. The mean direction is $\text{atan2}(\bar{b}_1, \bar{a}_1)$ from the first-harmonic moments of the pressure–velocity cross-spectra, weighted by S_η over the sea–swell band.

Near-bed velocity, stress and moments. Each bin of the velocity FFT is scaled from the sensor to the bed by $1/\cosh(kz_s)$ (linear theory, $u \propto \cosh kz$ with z up from the bed) and inverse transformed; U_b is the rms of $\sqrt{u_{\text{bed}}^2 + v_{\text{bed}}^2}$ and A_w the orbital excursion. Earlier revisions of this guide gave the factor as $\cosh(kH)/\cosh[k(H - z_s)]$, which the code's own header records as the original, wrong form. Wave bed shear stress uses the Swart (1974) friction factor with T_p, and roughness

$k_s = 2.5\,D_{84}$ where the site has a measured grain-size distribution (`shared/site_grain_size.m`), otherwise the legacy $k_s = 10\,D_{50}$ with the placeholder $D_{50} = 0.25$ mm. In catalog 2026-10-14.1, 19 of 65 records use a measured distribution, 4 an assumed site default (217.4 μm) and 42 the placeholder. Reynolds stresses and turbulent kinetic energy come from the detrended velocities. Velocity skewness and asymmetry (Hilbert form, negative for the forward-pitched shoaling shape) are in the `vmom` sub-struct.

8.3 L3: wave forcing characterization

Bands and storms. S_η is integrated over infragravity (0.004–0.04 Hz), swell (0.04–0.10 Hz) and sea (0.10–0.25 Hz) bands, with a dominant-band flag by energy flux. Storms are exceedances of $H_s > 1.5$ m lasting at least 6 h, merged across gaps under 12 h (fixed in `PUV_L3_storms.m`). Hourly CDIP MOP output for a window 7 days beyond each end of the record flags storms that fell in PUV data gaps.

Transport proxies. Bottom energy flux $F_b = \rho g \int c_g S_\eta \, df$ over the sea–swell band, and its running sum. Shields parameter $\theta = \tau_b/[(\rho_s - \rho)gD_{50}]$ from the L2 wave stress, with the Soulsby–Whitehouse critical value and a mobilization flag. Kinematic viscosity comes from each segment's mean temperature (`shared/seawater_nu.m`, with `Tvalid` screening the temperature), and settling velocity from Ferguson and Church (2004) with nominal-diameter constants $C_1 = 20$, $C_2 = 1.1$. The Rouse number uses the mean combined wave–current stress τ_m (Soulsby 1997; since 2026-08-13, before which the wave-stress amplitude overstated suspension), and is also given at D_{16}, D_{50} and D_{84} where a measured distribution exists. Six settling-velocity formulas could not be distinguished against the TBR23 surveys; they spread about $\pm 15\%$ at 250 μm.

Tides and currents. The tidal depth reference is the NOAA prediction for the config's `cfg.tideStation` (`shared/noaa_tide_predictions.m`); every San Diego config uses 9410230 (La Jolla, Scripps Pier). A config with no station gets 9410230 only when its instruments lie in San Diego County; elsewhere L3 stops with an error. Catalina (CAT21A/B) uses 9410079 (Avalon), 1.9 km from the instrument; before 2026-10-16 it used 9410230, which differs from Avalon by 2.5 cm rms over the CAT21A record. The station used is stored at `L3.tidal.station` and in the provenance, and the push refuses an L3 whose station no longer matches its config. Tidal harmonic analysis (`t_tide`, complex $u + iv$, seeded with `rng(0,'twister')` so the bootstrap is reproducible) runs on the segment-mean velocities over the span where velocity exists. Subtidal residuals (observed minus tidal) carry the wave-driven mean flow, including undertow. A failed NOAA download or a missing `t_tide` is an error unless `opts.allowTtideFallback` or `opts.allowNoTides` is set.

8.4 L4: nonlinear and infragravity diagnostics

`PUV_L4_eta` band-passes pressure-derived surface elevation (total, swell, infragravity); `PUV_L4_reflection` splits infragravity motion into incident and reflected parts (Sheremet method); `PUV_L4_boundwave` separates bound from free infragravity with the Hasselmann second-order kernel, and `PUV_L4_reflection_free` repeats the split on the free part, which gives the shoreline reflection that `L4.ref` alone overstates. The second-order theory over-predicts the bound part above $H_s/h \approx 0.10$, so filter on H_s/h before using `bound_frac`. `PUV_L4_bispectra` gives bicoherence, skewness and asymmetry ($K = 6$ sub-segments per hour, 60 degrees of freedom, 95% bicoherence level 0.316); `PUV_L4_moments` and `PUV_L4_velocity_pdf` give frequency-resolved moment correlations and pooled velocity histograms. Full field lists are in `docs/pipeline_levels.md`.

9 Validation

- **Against the MOP wave model.** PUV wave heights were compared with CDIP MOP spectra shoaled to the instrument depth. Analyses of April–July 2026 found the systematic difference organized by wave nonlinearity (H_s/h, Ursell number): the PUV has extra energy near the second harmonic of the peak that the linear model does not carry, with bound infragravity waves as its difference-frequency counterpart. A separate, real difference in directional spread does not explain the energy difference. An earlier conclusion that the model broadens the spectral peak was retracted in July 2026 as an artifact of comparing spectra on mismatched frequency grids. These results were computed on the catalog as it stood in July and are recorded, with the retractions, in `docs/compile_sections.pdf`; current versions are maintained with the paper that uses them.

- **Against an independent prior pipeline.** On Ruby2D MOP582_6m (Torrey Pines, October 2021 to February 2022; 2,322 matched hours, April 2026, 17-minute segments on the new side) the new pipeline reproduced the legacy archived L2 with R^2 of 0.98 for H_s (5 cm rms), 0.93 for mean direction, 0.88 for directional spread and 0.76 for T_p. Each fault of the legacy code measured in isolation, and which legacy products are safe to reuse, is in `docs/legacy_pipeline_defects.pdf`: H_s and mean direction are usable, directional spread and b_2 (hence radiation stress) are not.

- **Mean cross-shore flow.** In May 2026 the segment-mean cross-shore velocity of 38 instruments was tested against real Stokes return flow, a constant calibration offset and random Doppler noise, and matched return flow in sign, magnitude and response to averaging length. That run predates the heading corrections of 2026-05-14 (one included record had its velocity sign reversed) and later rebuilds, and has not been repeated.

- **Binary decoder.** Besides the bit-exact match to ASCII (§4), the binary-decoded deployments' hourly H_s was compared with CDIP MOP in July 2026: R^2 of 0.88–0.94 with 1–4 cm bias for the modern 2 Hz records (TOR19W, TOR20W, IB19W), and 0.5–0.9 for the firmware-1.21 records, which sit farther from their MOP transects (`docs/recopied_data_backlog.md`). Mean vertical velocity near zero in every decoded record supports the decode and the rotation.

- **Tests.** `run_all_tests` runs 23 self-contained tests, including synthetic closure tests of the pressure correction, the Z-test, channel QC, provenance, the tide station, the gap fill and the known-failure handling.

10 Output reference

L1 (`PUV` struct). `time` (UTC, stored without a time zone), `fs`, `P` (dbar), `BuoyCoord.{U,V,W}` (buoy-frame velocity), `T`, `doffp`, `rotation` (declination applied, coordinate system), and `PUV.qc` (`valid_vel`, `valid_p`, `valid_tilt`, `valid_T`, `vel_c_factor`, `vel_c_corrected`, `vel_rotation_static`, `opts`).

L2 (`L2` struct, one entry per segment). `time`, `Hs`, `Hs_SS`, `Hs_IG`, `Tp`, `Tm02`, `meanDir`, `Ef`, `depth`, `Ub`, `tau_b`, `a1`/`b1`/`a2`/`b2`, `S_eta`, the `vmom` sub-struct (`skewness`, `asymmetry`, `u_uabs2`, ...), `uMean`/`vMean`, `Tmean`, `shorenormal`, and the flags of §6.

L3 and L4. L3 holds the band, storm, transport (`shields`, `mobilized`, `Fb`, `rouse`, `ws`, `nu`, `tau_m`) and tidal (`tidal.depth_pred`, `tidal.station`, `tidal.depth_source`) fields; L4 holds `eta`, `ref`, `bispectra`, `boundwave`, `reflection_free`, `moments` and `pdf`. `docs/pipeline_levels.md` gives the full struct shapes.

Conventions. After L1: $+x$ west, $+y$ north, $+z$ up, true north. After L2 rotation: $+x$ onshore, $+y$ alongshore north; onshore flux is $q_x > 0$. Pressure in dbar, depths in m, times UTC. Config headings are magnetic; the declination is added at L1.

11 Known limitations

- **Disputed headings.** 12 records keep a compass heading more than 10° from the logged mount heading (`shared/heading_disputed_records.m`); their directional quantities are uncertain by at least that much.
- **Grain size.** 42 of 65 records use the placeholder D_{50}; bed stress, Shields, mobilization, settling velocity and Rouse on those records will move when measured values arrive. L4 does not depend on D_{50}.
- **Fixed `doffp`.** The bed can move tens of centimeters during a deployment (logged changes up to 23 cm). That moves depth and depth-normalized quantities (H_s/h, Ursell) by about 3% and bed stress by under 0.3%. A PUV cannot measure its own `doffp`. IB18W and IB19S carry values whose source is unknown.
- **Off-bed pressure QC is off** until the next deliberate L1 rebuild; no L2 segment in the catalog is affected.
- **Unsourced geometry.** 69 `doffp`/`latlon`/`shorenormal` values had no on-line source at the last audit (2026-07-28); treat them as unverified.

12 Troubleshooting and where to look next

- **A driver ends with an error.** Read the list it prints: configs that threw and records not written. Known field failures are skipped and do not cause it.
- **A config change seems to have no effect, or a deployment vanished from a batch.** Run `validation/audit_registry_loads.m`; batch loops skip a config that throws.
- **L3 stops on tides.** Check `t_tide` is on the path, `cfg.tideStation` is set, and NOAA is reachable.
- **Network reads are slow.** Cache raw files with `copy_raw_to_local(cfg)`.
- **Little velocity survives a storm.** Check the failure signature (§6.6): a sensor-block failure is recoverable, a topple is not. Confirm `Tvalid` is set to the site range.
- **Recovered (`qc_flag` = 3) data in an analysis.** Filter to `qc_flag` = 1 for clean-only, or include `qc_flag` = 3 knowing the scale uncertainty.

Further reading: `README.md` (current requirements, provenance, catalog), `CHANGELOG.md`, `docs/NEW_DEPLOYMENT.md`, `docs/pipeline_levels.md` (status and struct shapes), `PIPELINE_NOTES.md` (design decisions and known issues), `docs/catalog_changes/` (what each rebuild changed), `docs/L1_sensor_block_failure_2026-07-09.md` and `docs/OUTSTANDING_channel_decoupling.md` (the channel-decoupling work), `config/CONFIG_REVIEW_NOTES.md` (per-deployment review items).

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